MIL-P-46547A
12 January 1972
SUPERSEDING
MIL-P-46547(MU)
23 June 1961

MILITARY SPECIFICATION

PROJECTILE, 20MM, DUMMY, M51A2

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Projectile for use in the assembly of Cartridge, 20mm, Dummy, M51A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F7258861 - Carton, for Interplant Shipment of Projectiles, 20mm, Assemblies and Metal Parts Assemblies (200)
D11075759 - Projectile, 20mm, Dummy, M51A2
IEL 11075759 - Index of Inspection Equipment Lists for: Projectile, 20mm, Dummy, M51A2
D7259136 - Shipping Container for Inert Projectile, 20mm, Assemblies and Metal Parts Assemblies (10,000)

FSC 1305

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The projectiles shall comply with all the requirements of Drawing (Dwg.) D11075759 and with all requirements of the drawings and specifications shown thereon.

3.2 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.2.1 Processing defects.-The projectile shall be free of defects such as dents, deep scratches, cuts, excessive surface roughness, metal flaws, burrs, sharp edges, and other forms of substandard workmanship.

3.2.2 Cleanliness.-The projectile shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 25 projectiles. The sample shall be manufactured using the same equipment, processes and procedures as will be used in regular production.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

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4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications will result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.1 Lot.-A lot shall consist of projectiles produced by one manufacturer in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

4.3.1.2 Lot identification.-Each projectile and each packed projectile lot shall be identified in accordance with applicable drawings supplemented as directed by the procuring activity.

4.3.2 Examination.-Examination for major and minor defects shall be performed as specified herein. Each projectile found to be defective shall be rejected.

4.3.2.1 Sampling and acceptance criteria.-The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.3.2.2.c) In addition, where MIL-STD-1235 is used sampling plans applicable to a class shall not be used for major defects. (see 6.2)

4.3.2.2 AQL's.-The following AQL's are assigned:

a. Individual defect characteristic basis.

Major defects	0.25%
Minor defects	0.40%

b. Defect characteristic, class basis.

Major defects	1.50%
Minor defects	2.50%

c. Individual defect characteristic associated with the class basis.

Major defects	0.40%

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4.3.3 Classification of defects 1/.-The classification of defects shall be as follows:

4.3.3.1 Projectile (see Dwg. D11075759).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Runout, rotating band with bourrelet	Gage
102. Runout, diameter, rear of rotating band with bourrelet	Gage
103. Runout, nose, with bourrelet	Gage
104. Diameter, bourrelet	Gage
105. Diameter, rear of rotating band	Gage
106. Diameter, rotating band	Gage
107. Total length	Gage
108. Length to rear of rotating band	Gage
109. Weight	Balance
110. Profile, crimp groove	Gage
111. Foreign matter	Visual
Minor:	
201. Profile of rotating band	Gage
202. Location of crimp groove	Gage
203. Surface finish improper	Visual 2/
204. Marking incomplete or unidentifiable	Visual

1/ MIL-STD-651 shall apply in defining and evaluating visual defects.

2/ Measurement of surface roughness shall apply where visual inspection indicates that the specified finish is exceeded.

4.3.4 Packaging and marking inspection.-Inspection for packing and marking shall be in accordance with MIL-STD-1169.

4.3.5 Inspection equipment.-Index of Inspection Equipment Lists IEL 11075759 identifies inspection equipment lists indicating the equipment required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to Inspection Equipment List for definition of coding format).

5. PREPARATION FOR DELIVERY

5.1 Packing, Level C.-Each lot of product shall be packed as specified by Dwg. F7258861 or Dwg. D7259136, as directed by the procuring activity.

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5.2 Marking and labeling.-Marking or labeling of packed cartons shall comply with applicable requirements of MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Place of inspection, if not at place of manufacture.

6.1.3 Packing and marking instructions. (see 5.1 and 5.2)

6.2 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitations, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

Custodian:

Army - MU
Air Force - 70

Preparing Activity:

Army - MU

Review Activities:

Army - MU
Air Force - 70

Project No. 1305-0689

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